

Aman Anwar

540-934-8570 | amans.place | aman06@vt.edu | [GitHub](https://github.com) | [LinkedIn](https://www.linkedin.com) | [Devpost](https://devpost.com)

EDUCATION

Virginia Tech

B.S. in Computer Science

May 2028

Blacksburg, VA

TECHNICAL SKILLS

Java, JavaScript, TypeScript, Python, SQL, Docker, Flask, Node.js, PyTorch, Scikit-Learn, Pandas, React.js, REST APIs, Express, MongoDB, PostgreSQL, Windows Command Line, Linux Command Line, Bash Scripting, Google Cloud Platform

EXPERIENCE

VTHacks

Technical Lead

Jul. 2025 - Present

Blacksburg, VA

- Lead a 5-member development team to build a real-time hackathon management system supporting 850+ active participants

NASA RAM @ IDOpt Lab

Researcher

Jul. 2025 - Present

Blacksburg, VA

- Contributing to NASA's Regional Air Mobility (RAM) program under the mentorship of Dr. Darshan Sarojini.
- Engineered Python models to analyze airfare data and predict fares across west Appalachian routes.
- Designed a highly performant web tool to explore predicted trips on a deck.gl based map.

IDPRO @ Virginia Tech

Research Project Lead

Jul. 2025 - Present

Blacksburg, VA

- Leading a team of 5 in cutting-edge spatial audio research, collaborating with an industry-leading sound engineer to explore automation of Spatial Audio(ambisonics).
- Developed an end-to-end audio processing pipeline integrating OpenAI Whisper, Facebook DEMUCS, and LLM fine-tuning on Apple MLX for intelligent sound separation and scene adaptation deployed on AWS Bedrock.
- Authored a successful proposal securing a \$1,000 ICAT SEAD Grant to support ongoing research.

Lumiq

Software Engineer Intern

May 2025 - Aug. 2025

Remote

- Built an end-to-end churn prediction pipeline using Python and Scikit-learn, achieving a 95% F1-score in forecasting customer churn.
- Performed extensive exploratory data analysis, feature engineering, and SMOTE based resampling on 100K+ customer records to correct class imbalance and enhance minority recall by 40%.

PROJECTS

PiggyBank | 3 Awards @ HackPrinceton 2025 | Knot API, Photon iMessageKit, Flask, React

Nov. 2025

- Built a financial agent that learns users' financial habits to predict users' next transactions to nudge them via iMessage to help develop better financial habits.
- Integrated Knot's Production TransactionLink with Photon's iMessage Kit to deliver real transactional insights and nudges using a custom prediction model deployed on DigitalOcean, powered by vector search in Snowflake.
- Awarded 2nd Place by Knot, Best use of DigitalOcean by MLH, and Honorable mention by Photon Framework.

Atlas | 2nd Place @ Marriott Codefest 2025 | React, Flask, Google Cloud Platform

Feb. 2025

- Built an infinite-canvas trip planner using React Flow, Flask, and MongoDB Atlas, enabling visual, non-linear travel organization with automatic state saving and real-time sync.
- Integrated AI-driven personalization through RAG and vector search, plus Google Places API for an accurate places feed and the Nano Banana model for intuitive room requests.
- Strengthened platform security via JWT authentication, bcrypt hashing, and hard-to-guess upload URLs, and delivered analytics dashboards helping hotels refine marketing and guest experience.

Fit Finder | Winner @ HooHacks25 | JavaScript, Perplexity, MongoDB, Python

MAR. 2025

- Built a visual outfit similarity search engine using vector embeddings from OpenCLIP (ChromaDB) and Vector search from scalable MongoDB Atlas, enabling personalized apparel discovery in user uploaded closet pictures.
- Integrated Google Gemini to generate outfit metadata and segments, boosting vector search accuracy by 2x.
- Increased visual product search accuracy by 50% by integrating Perplexity Sonar with Gemini generated image descriptions.

AWARDS & RESEARCH GRANTS

1st Place Overall | 2025 CMDA Fall Data Competition

Nov. 2025

Academy of Data Science, Virginia Tech, Blacksburg, VA

Roger & Debbie West ICAT SEAD Grant

OCT. 2025

Virginia Tech Institute for Creativity, Arts, and Technology, Blacksburg, VA